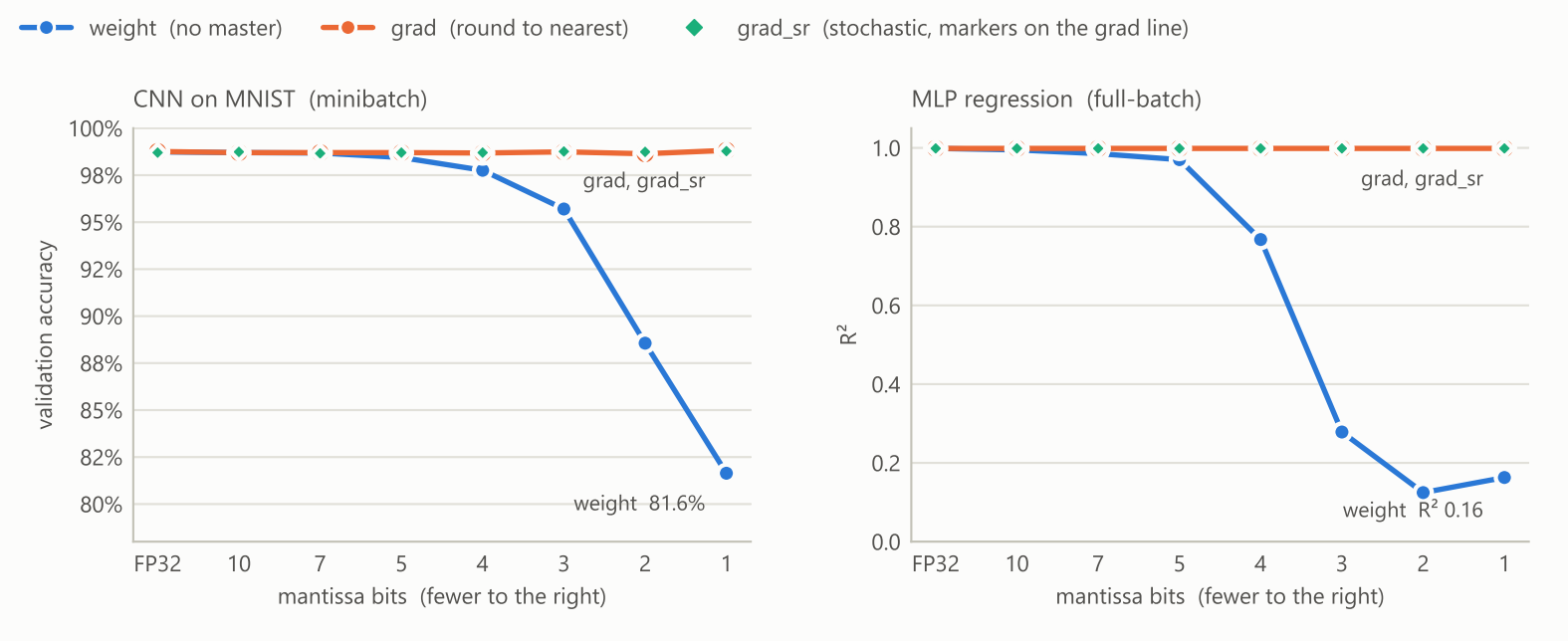
weight (no master)
grad (round to nearest)
grad_sr (stochastic, markers on the grad line)
CNN on MNIST (minibatch)
validation accuracy
100%
98%
95%
92%
90%
88%
85%
82%
80%
grad, grad_sr
weight 81.6%
FP32
10
7
5
4
3
2
1
mantissa bits (fewer to the right)
MLP regression (full-batch)
R²
1.0
0.8
0.6
0.4
0.2
0.0
grad, grad_sr
weight R² 0.16
FP32
10
7
5
4
3
2
1
mantissa bits (fewer to the right)